Library Management API

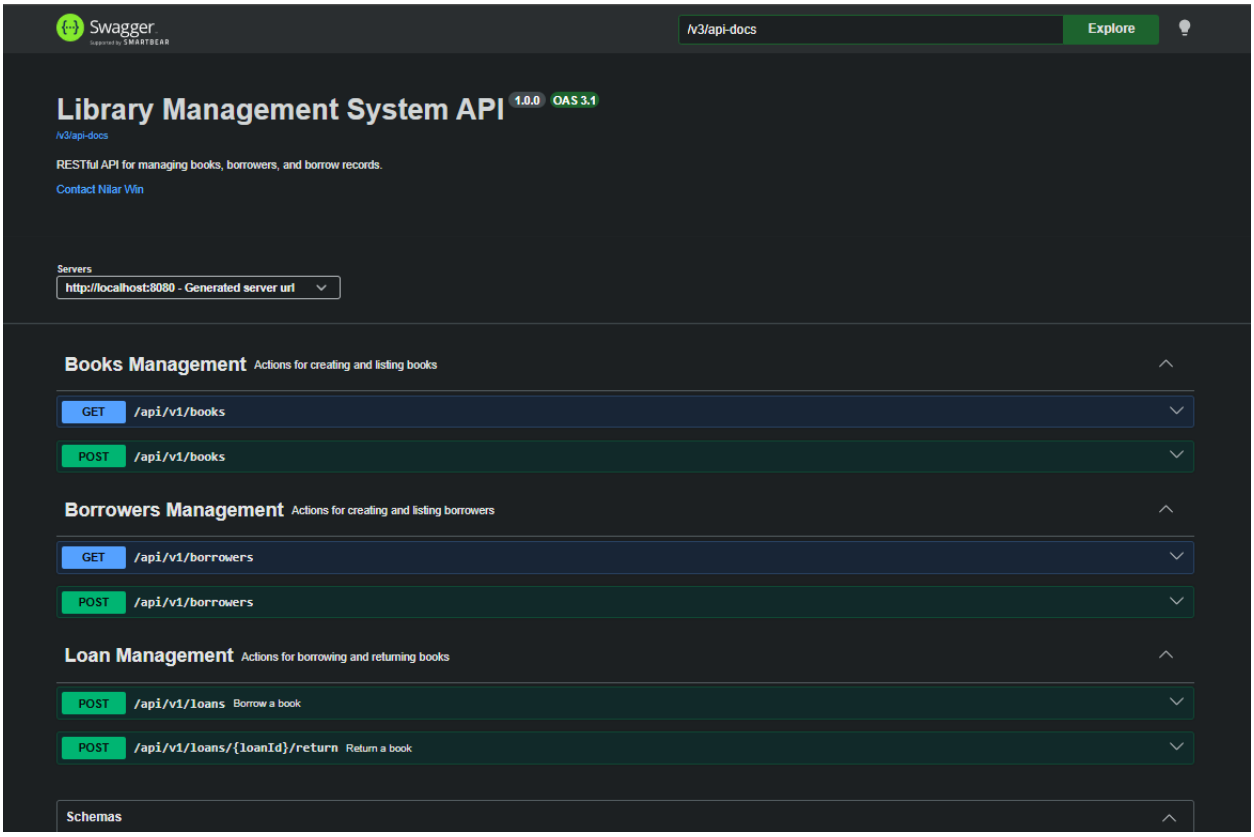
GitHub Repository Link: https://github.com/RebeccaNila/library-management

Build And Run

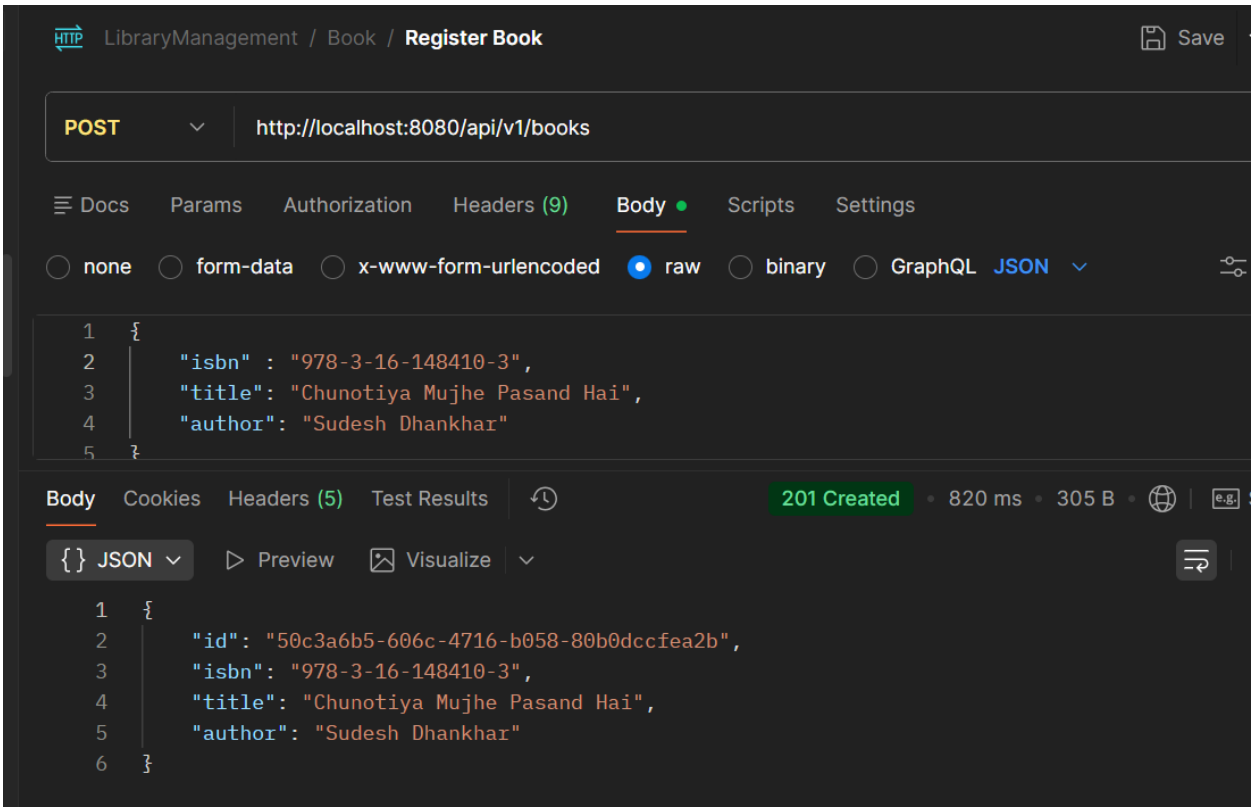
```
[INFO] Scanning for projects...
[WARNING]
[WARNING] Some problems were encountered while building the effective model for com:library-management:jar:0.0.1-SNAPSHOT
[WARNING] 'build.plugins.plugin.version' for org.jacoco:jacoco-maven-plugin is missing. @ line 117, column 21
[WARNING]
[WARNING] It is highly recommended to fix these problems because they threaten the stability of your build.
[WARNING]
[WARNING] For this reason, future Maven versions might no longer support building such malformed projects.
[WARNING]
[INFO]
[INFO] -----------------------< com:library-management >-----------------------
[INFO] Building library-management 0.0.1-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] >>> spring-boot-maven-plugin:3.5.12:run (default-cli) > test-compile @ library-management >>>
[INFO]
[INFO] --- jacoco-maven-plugin:0.8.14:prepare-agent (default) @ library-management ---
[INFO] argLine set to -javaagent:D:\\mavenRepository\\org\\jacoco\\org.jacoco.agent\\0.8.14\\org.jacoco.agent-0.8.14-runtime.jar=destfile=D:
Projects\\java\\intelliJ\\library-management\\target\\jacoco.exec
[INFO]
[INFO] --- maven-resources-plugin:3.3.1:resources (default-resources) @ library-management ---
[INFO] Copying 3 resources from src\main\resources to target\classes
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO]
[INFO] --- maven-compiler-plugin:3.14.1:compile (default-compile) @ library-management ---
[INFO] Nothing to compile - all classes are up to date.
```

Swagger

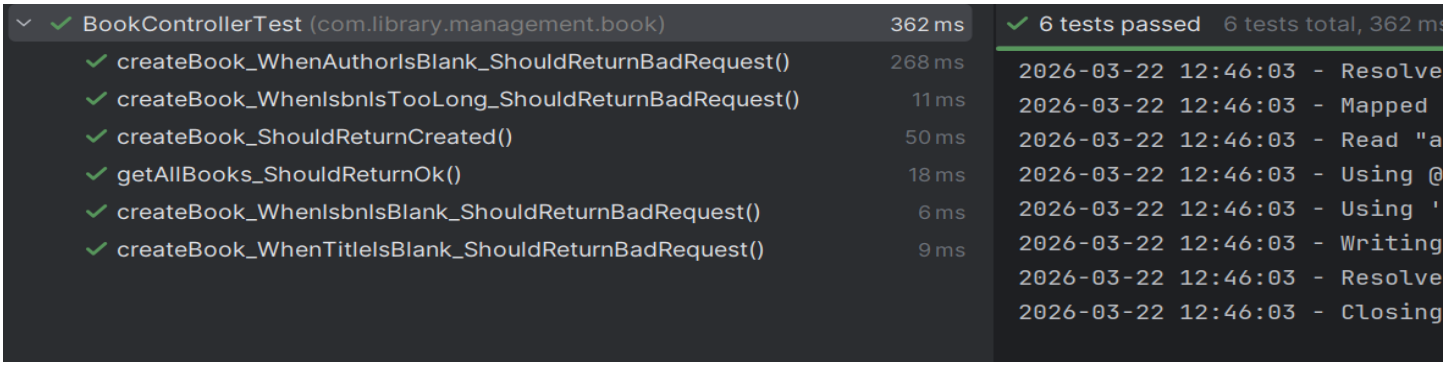
[Swagger UI](#)

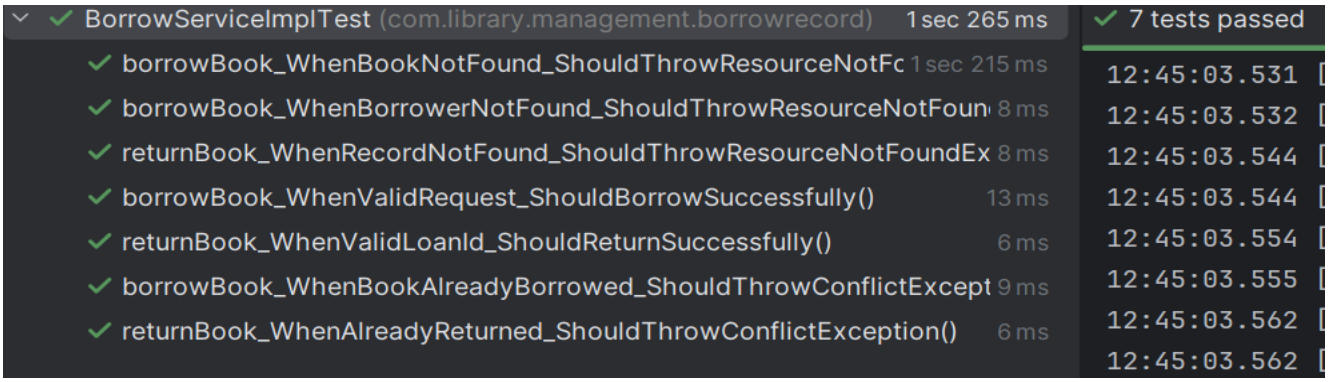


Postman Collection



Unit Test





Test Coverage

Generate the unit test coverage using JaCoCo (Java Code Coverage)

mvn jacoco:prepare-agent test jacoco:report

Folder to look report: target/site/jacoco/index.html

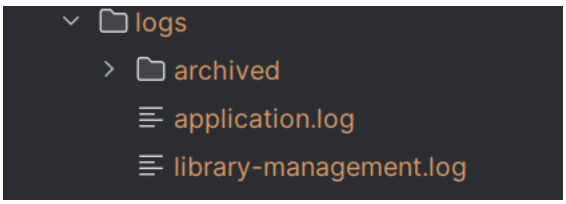
library-management

library-management

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
com.library.management.common.exception		60%		n/a	4	10	14	27	4	10	2	6
com.library.management		37%		n/a	1	2	2	3	1	2	0	1
com.library.management.borrowrecord		98%		100%	1	14	2	45	1	12	1	6
com.library.management.book		97%		100%	1	16	2	39	1	11	1	5
com.library.management.borrower		96%		n/a	1	10	2	29	1	10	1	5
com.library.management.common.config		100%		n/a	0	2	0	9	0	2	0	1
Total	55 of 672	91%	0 of 14	100%	8	54	22	152	8	47	5	24

Application Log

Project Root folder: \logs\library-management.log




```
2026-03-22 05:00:48.883 [main] INFO  c.l.m.LibraryManagementApplication - Starting LibraryManagementApp
2026-03-22 05:00:48.891 [main] INFO  c.l.m.LibraryManagementApplication - The following 1 profile is ac
2026-03-22 05:00:51.139 [main] INFO  o.s.d.r.c.RepositoryConfigurationDelegate - Bootstrapping Spring D
2026-03-22 05:00:51.303 [main] INFO  o.s.d.r.c.RepositoryConfigurationDelegate - Finished Spring Data r
2026-03-22 05:00:53.269 [main] INFO  o.s.b.w.e.tomcat.TomcatWebServer - Tomcat initialized with port 80
2026-03-22 05:00:53.364 [main] INFO  o.a.coyote.http11.Http11NioProtocol - Initializing ProtocolHandler
2026-03-22 05:00:53.373 [main] INFO  o.a.catalina.core.StandardService - Starting service [Tomcat]
2026-03-22 05:00:53.375 [main] INFO  o.a.catalina.core.StandardEngine - Starting Servlet engine: [Apach
2026-03-22 05:00:53.467 [main] INFO  o.a.c.c.C.[Tomcat].[localhost].[/] - Initializing Spring embedded
2026-03-22 05:00:53.469 [main] INFO  o.s.b.w.s.c.ServletWebServerApplicationContext - Root WebApplicati
2026-03-22 05:00:54.063 [main] INFO  o.h.jpa.internal.util.LogHelper - HHH000204: Processing Persistenc
2026-03-22 05:00:54.278 [main] INFO  org.hibernate.Version - HHH000412: Hibernate ORM core version 6.6.
```

Docker

Build and Run using docker compose

docker-compose -f deployment/docker/docker-compose.yml --env-file .env up –build -d

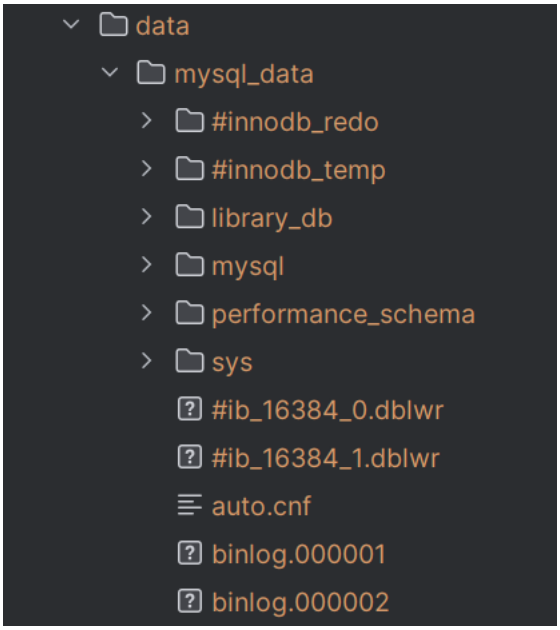
Docker Image

library-management-api	latest	007aeb11f1fa	1 minute ago	366.3

Docker Container

Name	Container ID	Image	Port(s)	Actions
library-management	-	-	-	
library_mysql	a6a514a7f791	mysql:8.0	3306:3306	
library_management_api	7279d616ffad	library-man	8080:8080	

Docker Date will store project root folder : \data\mysql-data\



Access the API : http://localhost:8080/api

Docker stop and remove the container

docker-compose -f deployment/docker/docker-compose.yml --env-file .env down

Kubernetes

Build Docker Image

docker build -t library-management-api:latest -f deployment/docker/Dockerfile .

Run k8 commands

Apply in order (first time)

kubectl apply -f deployment/k8s/namespace.yml
kubectl apply -f deployment/k8s/configmap.yml
kubectl apply -f deployment/k8s/secret.yml
kubectl apply -f deployment/k8s/mysql-pvc.yml
kubectl apply -f deployment/k8s/mysql-deployment.yml
kubectl apply -f deployment/k8s/mysql-service.yml
kubectl apply -f deployment/k8s/api-deployment.yml
kubectl apply -f deployment/k8s/api-service.yml

Verify Everything Running

```
NAME                                          READY   STATUS    RESTARTS   AGE
pod/library-api-deployment-7b97bd848-66tpw    1/1     Running   0          61s
pod/mysql-deployment-5765556fb4-8swn4         1/1     Running   0          75s

NAME                           TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
service/library-api-service    NodePort    10.101.79.70     <none>        8080:30080/TCP   57s
service/mysql-service          ClusterIP   10.110.243.250   <none>        3306/TCP         64s

NAME                                     READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/library-api-deployment   1/1     1            1           61s
deployment.apps/mysql-deployment         1/1     1            1           75s

NAME                                                DESIRED   CURRENT   READY   AGE
replicaset.apps/library-api-deployment-7b97bd848    1         1         1       61s
replicaset.apps/mysql-deployment-5765556fb4         1         1         1       75s
```

Check logs
kubectl logs -f deployment/library-api-deployment -n library-management

Access the API : http://localhost:8080/api

Kubenertest stop

```
# Delete everything including PVC (wipes DB data)
kubectl delete all --all -n library-management
kubectl delete pvc --all -n library-management
```